

HAZEM MASARANI

Software Engineer · Distributed Systems · ML Infrastructure

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PROFESSIONAL SUMMARY

Software engineer and published ML-systems researcher experienced in Python automation, data-intensive backend platforms, algorithms, and multi-GPU inference. M.S. Data Science candidate (3.96 GPA) who has advanced from junior to senior engineer and delivered systems supporting live telecom operations. **TECHNICAL SKILLS**

Languages: Python, Java, C/C++, JavaScript, SQL, R **Systems:** Distributed systems, parallel computing, Linux, Docker, Git, APIs

Frameworks/Data: PyTorch, TensorFlow, Hugging Face, Pandas, NumPy **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB **EXPERIENCE**

PACELab, Stony Brook University

Graduate Researcher | Advisor: Prof. Anshul Gandhi

Aug 2025–Present

Stony Brook, NY

- Build tensor-parallel inference for Mamba-family state-space models, sharding projections and SSM parameters across multiple GPUs.
- Validate distributed execution against single-GPU reference outputs and profile throughput, latency, memory, and communication overhead.
- Co-authored a communication-efficient design using cross-phase state caching, locality-preserving partitioning, and quantized AllReduce.

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Senior Data Automation & Network Data Engineer

Jan 2024–Sep 2024

Egypt | Hybrid

- Led Python automation and multi-source platform development integrating alarms, topology, rollout, availability, and maintenance data.
- Built live operational modules and deployed a Dockerized OpenStreetMap tile server for real-time geospatial network visualization.
- Developed root-cause and financial-impact logic used to prioritize repairs and infrastructure investment; mentored engineers in Python, SQL, and algorithms.

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Junior Data Automation & Network Data Engineer

Jul 2023–Jan 2024

Egypt | Hybrid

- Integrated heterogeneous sources into the eSW platform and developed a chatbot for internal data access and recurring analysis.

SELECTED ENGINEERING PROJECTS

Tensor Parallelism for Mamba2 | GitHub

Distributed Mamba2 inference with sharded mixer/SSM parameters, output validation, and repeatable multi-GPU benchmarks.

FM-Index Engine | GitHub

C++/Python pipeline for linear-time DC3 suffix arrays, Burrows–Wheeler Transform, FM-index construction, and backward search.

DC3 / Skew Benchmark Suite | GitHub

Benchmarked naive, doubling, and DC3 suffix-array algorithms for runtime and memory in C++ and Python. **PUBLICATION**

Scaling State-Space Models on Multiple GPUs with Tensor Parallelism | arXiv:2602.21144

Dutt, Shah, Masarani, Gandhi · February 2026 **EDUCATION**

Stony Brook University — M.S. Data Science, GPA: 3.96/4.00

Expected Dec 2026

Alexandria University — B.Sc. Computer & Communication Engineering, Honors, GPA: 3.47/4.00

2022